DATA & DECISIONS

EPA and Coal Plants

Data & Decisions · Group Assignment 1 · 100 points

Your team is preparing a national emissions summary. Compare two sampling designs and determine how to estimate average emissions when states contribute different numbers of coal-fired generating units. Explain your reasoning, not just your results.

Data and analysis frame

Use `EPA_CoalUnitCharacteristics2022.xls`, the unchanged EPA 2022 data supplied in bCourses. Analyze 2022 SO2 (tons), 2022 NOX (tons), and 2022 CO2 (tons). Each data row represents a generating unit; a plant may contain several units. Keep the unit-level rows rather than aggregating by plant.

Use one analysis frame throughout: units with reported values for all three pollutants. Exclude units with any missing pollutant value, retain reported zeros, and do not replace missing values with zero. Record the excluded unit identifiers and reason. The estimand for your calculations is average annual emissions per unit in this analysis frame, not emissions per plant or per MWh.

You may use Excel, Stata, R, Python, or another transparent analytical tool. AI assistance is permitted but optional. Use sampling without replacement. Record a random seed when supported; otherwise preserve the random values and selected unit identifiers so the submitted sample stays fixed and can be checked.

1. Define the frame and calculate benchmarks (20 points)

a. Report the number of rows before and after the missing-data rule, the number of distinct plants and represented states in the analysis frame, and the number of units in each state. Use the plant identifier `ORISPL` to count plants and `ORISPL` plus `Boiler ID` to track sampled units; check that the combined unit identifiers are unique.

b. Calculate the full-frame mean of each pollutant for the United States and for Arizona. Label all means with their units.

c. Explain which population these benchmarks describe. Does having every row in this file establish coverage of every US coal-fired unit? Explain what the file and the exclusions do, and do not, establish. You need not conduct new external research.

2. Draw a national simple random sample (20 points)

a. Draw a simple random sample of 150 units from the analysis frame. Save the selected identifiers and sampling method. Check that it contains exactly 150 distinct eligible units.

b. Calculate each pollutant mean in your sample. Present the sample means beside the national benchmarks from Part 1 and discuss the differences. Does a valid random sample have to reproduce the benchmarks exactly?

c. Report how many Arizona units were selected. Calculate their pollutant means if any were selected; otherwise report that no Arizona mean can be calculated. Compare with the Arizona benchmarks and explain why national random sampling may provide limited subgroup information.

3. Sample within states (20 points)

a. Independently draw a random sample within each state represented in the analysis frame: five units without replacement, or all units if the state has fewer than five. Record the sampling method and selected identifiers. Save a table containing each state's full-frame count and selected count.

b. Check that each state has the required number of distinct eligible units. Calculate the Arizona sample means and compare them with the Arizona benchmarks and the Arizona results from Part 2.

c. Explain what this design guarantees about Arizona coverage. Does it guarantee that the Arizona means from this particular sample will be closer to the benchmarks? Distinguish guaranteed coverage from sampling variation.

4. Estimate national means and verify your analysis (40 points)

Includes 10 points for the verification in part d.

a. Pool the units selected in Part 3 and calculate the unweighted mean of each pollutant. Compare these estimates with the national benchmarks. Explain why covering every represented state does not by itself make the pooled mean appropriate for the national unit-level estimand.

b. First calculate this small example by hand: State A has 100 units and a sample mean of 20 tons; State B has 10 units and a sample mean of 80 tons. Five units were sampled in each state. Calculate the unweighted pooled mean and an appropriately weighted national mean. Then construct weights for the EPA state samples that reflect each state's share of units in the full analysis frame. Calculate a weighted national mean for each pollutant. Show your formula and a state-level table with full-frame counts, sample counts, sample means, and weights. Explain the treatment of states with fewer than five units. Verify that your normalized weights sum to one.

c. Compare the full-frame benchmark, national SRS estimate, unweighted stratified estimate, and weighted stratified estimate in one table. Explain what the weighting corrects and why a correctly weighted estimate may still differ from the benchmark. Do not infer that one design is always more accurate from a single draw.

d. Identify one consequential analytical choice. Show how you checked it, and explain whether it changed your conclusion. Examples include the unit of analysis, missing-data rule, sampling without replacement, state counts, or weighting. Check using a separate calculation, formula inspection, selected-row inspection, or another transparent method. Confirming a correct analysis counts; you do not have to find an error. Simply asking the same AI to confirm its answer is not sufficient.

Submission and team responsibility

Submit one completed assignment per study team through bCourses by the posted due date. Include all contributing members' names. One member uploads on behalf of the team; all members must contribute and understand the work. Work within your team rather than consulting other teams or outside people.

Include (1) concise written answers with clearly labeled tables; (2) a reproducible analytical file, such as a spreadsheet with formulas and saved samples, or code with its required input and seed; and (3) brief provenance for material AI use, if any. Supply a shareable chat/export when supported or the relevant prompts and outputs, plus a short statement of what AI helped with and what you verified, modified, or rejected. If no AI was used, say so. No screenshots of every interaction are required.

Grading rewards correct calculations, clear reasoning, reproducible implementation, and the verification check. The four parts total 100 points. Be prepared to explain sampling and perform a small weighted-average calculation without AI or software on the final exam.